

Lesson 125 (0.5) Ground (0.8) Local Dual (0.8) Local Solo

Lesson Objective

The purpose of this lesson is for the applicant to conduct their second solo flight. Prior to the solo portion of this lesson, a dual flight is conducted with instruction on pattern entry and departure procedures to ensure the applicant is ready to solo under the current conditions..

Briefing

- Outline the objectives and completion standards of the lesson.
- Establish the student and instructor responsibilities.
- Review homework assigned in the previous lesson.
 - Review relevant airport diagrams with the student.
- Student questions.

References

- Airplane Flying Handbook. ([FAA-H-8083-3C](#))
 - Chapter 06 - Takeoffs and Departure Climbs.
 - Chapter 08 - Airport Traffic Patterns.
 - Chapter 09 - Approaches and Landings.
- Pilot's Handbook of Aeronautical Knowledge. ([FAA-H-8083-25B](#))
 - Chapter 14 - Airport Operations.

Instructor Guidance

- This is the student's second solo flight. The dual portion should be used to assess their proficiency under today's specific conditions — wind, traffic, and runway in use — before making the decision to send them solo. Do not treat the dual portion as a formality simply because they have already soloed once. Every solo endorsement decision must be made fresh based on current performance.
- Apply the same standard as Lesson 124 — the student must demonstrate three consecutive safe takeoffs, traffic patterns, and landings without instructor input before the solo portion begins.
- Before departing, brief the student thoroughly on the airport diagram for the runways in use. Quiz them on expected taxi routes, hold short points, radio calls, and the instructor drop-off and pickup procedures — whether at the KMLB north t-hangars or at the X59 FBO. A student who is unsure of the taxi routing to retrieve you after the solo should not begin the solo until that is resolved.
- If operating at KMLB, brief the specific exit runway and ramp entry procedure clearly
- Borrow a handheld radio from the line department and monitor the student's radio calls throughout the solo. Remain reachable and trackable via Flightradar24 or FlightAware for the duration of the flight.

- Remind the student to announce "student pilot solo" on initial contact with Melbourne Ground and Melbourne Tower. This is a safety practice that ensures ATC is aware of the student's status and can provide additional support if needed.
- During the dual portion, focus the pattern entry and departure procedures — this is specifically called out in the lesson objective and should be the instructional emphasis of the dual flight, in addition to confirming overall proficiency.
- Include at least one deliberate go-around during the dual portion. Confirm the procedure is executed correctly and confidently before the solo begins. A hesitant or incorrect go-around on the dual portion is a signal to delay the solo.
- Use the post-brief to give the student specific, constructive feedback on their solo performance. Ask them what they felt went well and what they would do differently — building self-assessment skills at this stage is just as important as the flying itself.

Lesson Content (Ground)

Supervised Solo Plan of Action

- Brief the plan of action to the student:
 - The first part of the flight will be a dual flight.
 - The student must demonstrate that they can conduct at least three safe takeoffs, traffic patterns, and landings.
 - No instructor guidance.
 - All landings must be full stop taxi backs.
 - Have the student take out their airport diagram.
 - Based on the runways in use, ask the student about what taxiways they should expect, radio calls, procedures, runway incursion avoidance, etc.
 - If the lesson is taking place at KMLB, the instructor will be getting dropped off at the north side, brief the student about the procedure.
 - Exit the runway at M or Q.
 - Enter the North t-hangar ramp at M or K1.
 - If entering at M, make the first right after entering the ramp.
 - If entering at K1, make the first left after entering the ramp.
 - Once exiting the aircraft, the student must call ground control to request taxi before blocking the M or K1 entry/exit.
 - Once the solo is completed, the student needs to request a full stop at the North t-hangars and enter as briefed above.
 - For guidance, refer to the airport diagram at the bottom of this lesson plan.
 - If the lesson is taking place at X59, the instructor will be dropped off at the FBO
 - Ensure the student has the updated airport diagram for X59
 - Ensure the student is familiar with the various taxi routes to the FBO

depending on the active runway

- The second part of the flight will be a solo flight.
 - As aforementioned, all landings must be full stop taxi backs.
 - The student should mention "student pilot solo" on initial contact with Melbourne ground and Melbourne tower.
 - Monitor the student's radio calls through the portable radio.
 - Track the student's pattern visually if able or through an app (Flightradar24, FlightAware, etc.)

Go/No-Go Decision

- The student must make a go/no-go decision depending on:
 - Personal minimums.
 - FIT minimums.
 - Aircraft.
 - Airspace.
- Review the students planning with a focus on:
 - Weather, weight and balance, performance, flight planning, and fuel calculations.
 - NOTAMs, TFRs, FIFs, aircraft documents, and personal documents.
 - Equipment requirements (e.g. flashlight, fuel card, drinking water, view-limiting devices, life jackets, fly away kit, survival kit, etc.)
 - FRAT/TEM/safety considerations for the flight.
- While this is represented by one line item prior to departure, the go/ no-go decision is continuous and should be a consideration even once airborne.

Lesson Content (Dual)

Checklist Procedures

- Ensure student is adhering to a Florida Tech Aviation standardized Checklist

Preflight Inspection

- Preflight is done with checklist in hand
- Identify fuel/oil levels, ensure required equipment & documents are onboard and overall aircraft is an airworthy condition.

Engine Starting

- Correct engine start procedure is being implemented

Operation of the Communications and Navigation Units

- Demonstrate proper radio setup and frequency changes
- Verify navigation aids are tuned, identified, and operational

Radio Communication Procedures and ATC Phraseology

- Standard phraseology for clearance and ground/tower communication
- Emphasize listening skills and correct readbacks

Visual Scanning and Collision Avoidance

- Use FAA-recommended scan patterns
- Apply "see-and-avoid" with emphasis on traffic pattern operations

Taxiing

- Maintain centerline control with appropriate
- Use wind correction controls during taxi in varying conditions

Normal and Crosswind Takeoff

- Align and maintain runway centerline with rudder inputs
- Apply crosswind corrections on takeoff roll and initial climb.

Traffic Pattern

- The student will be in charge of all of the radio calls with tower and ground.
 - Emphasize the use of saying "say again" if the instructions are not clear.
 - Emphasize the use of saying "unable" if the instructions are not safe or cannot be followed.
- Traffic patterns are to be completed by the student without instructor guidance.

Go-Arounds or Rejected Landings

- Have the student conduct a go-around during one of the traffic patterns.
- Ensure that the correct procedure is conducted

Normal and Crosswind Landings

- Verify correct airspeeds.
- Verify timing of the round out and the flare.
- At least one main gear should be on the centerline.
- Longitudinal axis of the airplane must be aligned with the runway, no side-loaded landings

Normal and Crosswind Landings

- Verify correct airspeeds.
- Verify timing of the round out and the flare.
- At least one main gear should be on the centerline.
- Longitudinal axis of the airplane must be aligned with the runway, no side-loaded landings.

Power Off Landing

- While on the downwind, bring the power to idle.
- Ensure that the student pitches for 73 knots.
- Guide the student through when to turn base/final depending on the wind (if needed).
- The student must be able to land on the first third of the runway.

Lesson Content (Solo)**Checklist Procedures**

- Student pilot solo

Engine Starting

- Student pilot solo

Operation of the Communications and Navigation Units

- Student pilot solo

Radio Communication Procedures and ATC Phraseology

- Student pilot solo

Visual Scanning and Collision Avoidance

- Student pilot solo

Taxiing

- Student pilot solo

Normal and Crosswind Takeoff

- Student pilot solo

Traffic Patterns

- Student pilot solo

Go-Arounds or Rejected Landings

- Student pilot solo

Normal and Crosswind Landings

- Student pilot solo

Postflight Procedures

- Student pilot solo

Post-Brief

- Student self-critique.
- Instructor critique.
- Questions from the student.
- Lesson grading.
- Logbook completion.
- Schedule the next lesson.
- Preview the next lesson → 126.

Completion Standards

This lesson is complete when the applicant can:

1. Complete at least three stop and go or full stop normal and crosswind takeoffs and landings without instructor guidance while dual.
2. Execute a local solo in accordance with all Florida Tech Aviation, FAA and local procedures and regulations.